AI DISRUPTION AND FALLING COSTS IN THEORY RESEARCH

A PREPRINT

Gabriel Manso[1,2] **Claudio Mayrink Verdun**[2] **Neil Thompson**[2]

[1]MIT EECS [2]MIT FutureTech
Correspondence: gmanso@mit.edu, neil_t@mit.edu

A theoretical paper in mathematics or a related quantitative science represents a costly investment of expert time. AI has the potential to revolutionize this type of knowledge production, but early models singularly failed to do so. Frontier models, today, however are claiming important discoveries, but some of these claims are contested. To *systematically* assess how AI's capabilities are changing, we compare how well model releases from December 2024 to April 2026 solve the cutting-edge theory questions by using cases where we have ground-truth human answers that were unavailable to AI models when answering. These answers come from 254 recent arXiv papers from seven scientific domains spanning mathematics, physics, and economics. We show that progress has been enormous. Over these sixteen months, the share of problems for which models reached the same conclusion as the human authors rose from 2.8% (o1) to 34.6% (GPT-5.5), an increase of 22 percentage points per year. Gains reach every subfield at similar rates. These results were achieved at minimal cost, approximately $5 per attempt or ($15 per success) for GPT-5.5. We interpret these results as signs of a rapidly evolving shift in science in which some areas become substantially more cost-effective, making powerful theoretical tools available to a wider range of scientists.

Introduction

Science as we know it today, with its division of labor, its methods, and its research questions, is built around human expertise being scarce. This creates costs and incentives to which the practice of science responds[1]. Tools that reduce the cost of a task change not only how quickly existing work is done but also the kind of work that gets done. For example, when a $150 video game peripheral made it possible to capture and process three-dimensional motion data, a task that previously required custom code and $10,000-$20,000 worth of cameras, researchers not only published more but were able to study questions farther away from their original research areas[2]. If theoretical reasoning becomes cheaper, science could be similarly transformed – with theory discoveries becoming commoditized. Understanding the pace and scope of such changes requires knowing how successful AI is at doing theoretical reasoning and how quickly those costs are falling. In the case of mathematical research, as Terence Tao recently said, *How should the mathematical community respond to the advent of modern AI technologies, and their real and/or claimed capabilities to perform mathematical tasks?...The answer to this question depends crucially on a subquestion about what AI tools will actually be able to do.*[3]

There are important - but contested - reasons to believe that this disruption to theory research is already underway. AI models can now succeed on graduate and research-level benchmarks in science and mathematics[4,5] and have been used to produce new mathematical results[6–9], including for some celebrated problems: a proof by Anthropic that more than two thirds of the nontrivial zeros of the Riemann zeta function are simple and lie on the critical line[10], a proposed proof by OpenAI of finite-time blow-up for the Navier–Stokes equations that awaits assessment and is already being scrutinized[11], and a new result in matrix multiplication[12]. Importantly, some AI-produced claims seem to be overblown, even in cases where they claim to be formally verified, because the verification may not match the natural-language generated proof[13]. To help address this type of concern, we present an alternative test of AI progress where the answer that is being generated by these models is compared against the result that human researchers derived for the same question in a recent arXiv preprint. These human results are not guaranteed to be correct, since most preprints have

not yet been peer reviewed, but they provide an independent, expert-produced reference that was posted after the models' knowledge cutoffs. They are also produced by researchers at institutions selected using model-generated lists documented as informed by QS subject rankings, Nature Index, ShanghaiRanking's Global Ranking of Academic Subjects, and, for economics, RePEc/IDEAS, alongside assessments of historical influence and research reputation in each subfield. This is, of course, an imperfect proxy for quality, but it provides an additional signal that the human derivations may be of high quality.

To estimate how capable AI is in theoretical research, we present it with the question and stated premises of various scientific papers from the arXiv repository and use a lightweight agentic pipeline built on publicly available API models to solve it. We apply this pipeline to 254 studies posted between December 2025 and May 2026 across seven subfields of mathematics, statistics, physics, and economics. The AI result is then compared to the human-produced result from the arxiv paper, to assess whether the AI answer is correct. Since checking several technical derivations across seven specialized fields by hand would require a team of domain experts, this comparison is performed by an LLM grader that scores each answer against a rubric derived from the authors' result. To check the grader, we also had a mathematician check a handful of examples and their general conclusion followed a common pattern. In their own words, "The derivation seems to be correct, but there is some notation and ways to present the arguments that could certainly be clearer. However, if I were to referee a paper with this content, I would approve the result, accept the paper, and recommend a careful rewrite of parts of it." Holding questions and grading fixed, we vary the AI model across six OpenAI releases, from o1 (December 2024) to GPT-5.5 (April 2026). These form the longest series of public reasoning models and underlie ChatGPT, the most widely used AI tool by researchers[14]. The share of answers receiving full marks rises from 3% to 35%, and derivation cost per full-mark answer falls from about $170 to $15. Even with all pipeline stages included, a back-of-the-envelope calculation suggests that the AI costs are less than 0.3% of what it would cost to have a human expert solve these problems, raising questions about how the work of theorists would change with this powerful new tool.

More generally, what happens to science when the price of doing some tasks falls? A first answer focuses on the effect if we hold the organization of research fixed. In this case, the gains from automating tasks in science[15] depend on their share of research costs and the savings achieved within them. In particular, the gains can be limited by non-automated "bottleneck" tasks that remain costly when tasks are strongly complementary[16]. For example, to be useful, a theoretical model might need empirical calibrations that require costly, time-consuming experiments.

If we also allow the organization of science to evolve, a second answer emerges. When some tasks become cheap to do, the structure of science may need to evolve to take advantage of it. Scientists are already reporting on this phenomenon. Even while saving nearly 7 hours per week from AI, scientists observe a growing backlog of untested hypotheses and substantial demand for output verification from AI models[17]. Under these conditions, we would expect the field to evolve so that more scientists are focusing on these other complementary tasks. Based on our results, we would expect similar adjustments within the theory community, as some of their historical tasks become much cheaper to do with AI.

For most fields, one important part of the process is verification, since it can require experiments with procedures such as cell cultures, chemical reactions, or clinical trials, all of which require money and time. Even in mathematics, where proof assistants, when combined with AI, can now check proofs at scale, one must still establish that the formal statement matches the intended claim[18,19] and that the result is not just AI producing a correct derivation of something else[20,21]. Indeed, this critique has been leveled at recent mathematical results released by top AI labs, which have sometimes failed to withstand the scrutiny of verification[13]; see, for example, *Proofs and Prompts* for the reactions of several mathematicians to OpenAI's October 2026 release of more than 700 AI-generated manuscripts, whose correctness, readability and attribution are still being assessed. Our study measures the cost of producing candidate derivations evaluated against human-produced references, with the benefits for research depending on the additional work needed to verify and use the answers.

1 Benchmarking theoretical reasoning

We develop an agentic system to answer theoretical research questions constructed from recent papers. While answering, the system cannot see the source paper or retrieve external information, but it can use local numerical and symbolic tools. Our fixed dataset consists of 254 research questions produced from first-version arXiv papers released between December 2025 and May 2026. Our dates are chosen to be after the knowledge cutoff among the models we evaluate.

Our protocol follows three sequential stages: *Curation*, *Derivation*, and *Evaluation* (Figure 1a, Extended Data Figure 2; Methods). *Curation*, the only stage with access to source papers, first evaluates whether the study's central scientific claim can be expressed as a clear question and whether the result can be derived from sufficient premises using reasoning and standard domain knowledge. Studies whose main contribution requires paper-specific numerical simulations, observational data, or physical experiments fall outside the scope of this work, but are explored in other work[22,23]. For

each "retained" paper that meets our inclusion criteria, GPT-5.5 formulates the question and premises that make it well-posed, along with grading material withheld during *Derivation* and used in *Evaluation*. *Curation* runs once per paper, so all models receive the same question and premises as their *public input*.

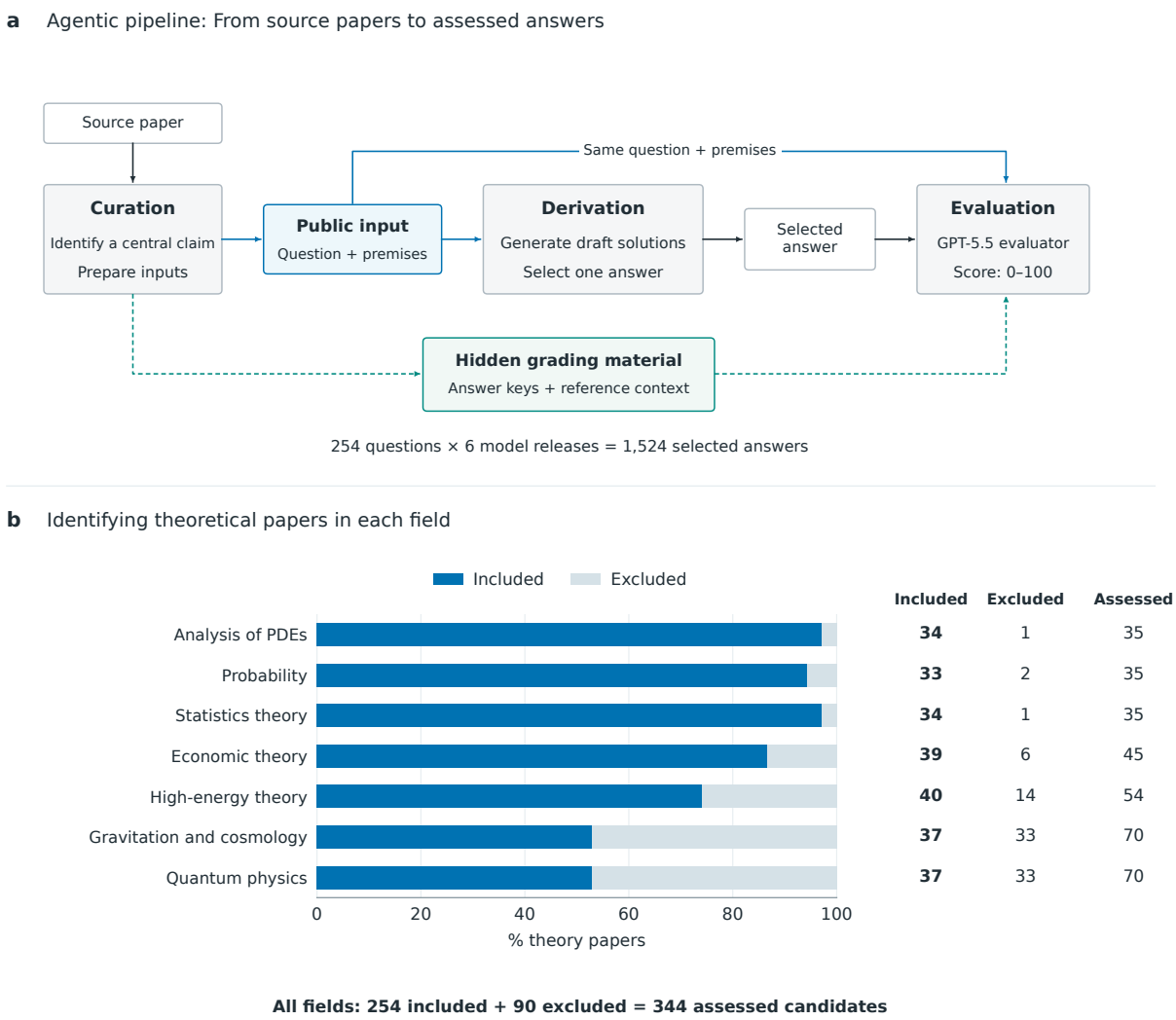


Figure 1: **Pipeline and benchmark construction. a**, GPT-5.5 performs Curation and Evaluation. Curation prepares the public input and hidden grading material; Derivation receives only the public input; Evaluation grades the selected answer using both. **b**, Percentages of assessed papers included and excluded under the curation criteria within each field. Bars sum to 100%, with counts on the right.

We intentionally design our study, so that our results are relevant not just to the exact papers that we test, but to the larger set of theoretical papers on arXiv. To do this, we randomly sample candidate papers from each field (with an affiliation-filter discussed in (Methods)), so that our results are representative, in expectation, of AI performance on all comparable papers within that frame. We determined which papers to include in our analysis by sampling within each subfield until we got 30-40 examples that could be included in the study (for statistical power reasons). In some areas, this required assessing fewer papers; in others, more. These differences reveal the fraction of papers in each field which are likely to be addressable by AI (and for which our estimator is relevant). In particular, 97.1% of the assessed papers in both analysis of partial differential equations (`math.AP`) and statistics theory (`math.ST`) passed our curation process (34/35 in each), compared with 52.9% in both gravitation and cosmology (`gr-qc`) and quantum physics (`quant-ph`) (37/70 in each; Figure 1b). These rates describe eligibility under our protocol, not how successfully models answer the resulting questions. Most included papers (237 of 254) concern mathematical statements, such as theorems, bounds, and closed-form expressions. The other 17 come from papers classified as *Methods & Resources* whose conclusions

can be derived mathematically. As an example, we included a paper on the quantum simulation of collective neutrino oscillations, because the system's symmetries can be used to show mathematically that the required quantum memory can be reduced[24]. On the other hand, we excluded a paper that introduces the Kaiwu-PyTorch plugin, because its central result requires empirical validation[25].

During *Derivation*, each AI model release receives the *public input* and attempts to reach the source paper's conclusion without internet access or external retrieval. Models can call numerical and symbolic tools to check calculations, test consistency, and search for counterexamples, as well as a theorem-hypothesis checklist that uses the evaluated model to assess whether the assumptions of invoked theorems are satisfied (Methods). The prompt instructs models to use these tools for specified types of claims, but the use of tools is not enforced. The pipeline requests three independently generated solution drafts, with agents prompted to reason, check, and decompose their work into claims. One draft is then selected using recorded verification and error indicators and passed to *Evaluation* (Methods). The same GPT-5.5 evaluator grades every model's selected solution using the *public input* and *hidden grading material*, without access to the source manuscript or tools. We provide a grading rubric and reference explanations constructed during *Curation* from the human-authored source paper, with each rubric requirement specifying the elements that the answer must satisfy. Element-level majority votes give an initial classification for each requirement: full credit if all elements are satisfied, partial credit if only some are satisfied, and no credit if none are satisfied. A conservative dissent rule can lower this classification (Methods). Hence, a score of 100% means that the selected answer receives full credit for all requirements and no penalties.

We emphasize that our results are specific to this agentic system and our limited computational budget. Other system designs (Supplementary Note S2), larger budgets, or a different evaluator could yield different results (Methods). Our estimates therefore describe what one documented pipeline achieved under its recorded configurations and costs, providing a reproducible reference point for tracking progress rather than a ceiling on what language models can do.

2 Performance gains and the cost of derivation

The percentage of answers receiving full marks increased from 7 of 254 for o1 (2.8%) to 88 of 254 for GPT-5.5 (34.6%), an increase of 31.9 percentage points between the first and latest evaluated releases (Figure 2a). Average performance followed a similar trend, rising from 29.9% to 69.1%, with a fitted linear increase of 2.36 score points per month (Figure 2b). Improvements were therefore seen both in mean scores (which allowed partial credit) and in the fraction of answers receiving full marks. A fractional logistic model, which constrains scores to lie between 0 and 100%, yielded a similar curve over the observed period (Methods).

As already mentioned, these answers are graded by an agentic system. As such, one might worry that the system could incorrectly categorize whether answers are correct, as has been seen in other work[26]. To help address this concern, we provide a grading rubric derived from the source paper. Previous research has shown that problem-specific grading rubrics can improve agreement between LLM and expert scores on competition-level mathematical proofs[27]. Nevertheless, we also ask a professional mathematician to review the AI results. Because of the large time commitment involved, this review is necessarily limited to five selected problems. The mathematician confirms the automated verdict on four of them and partially agrees with the fifth, noting that some justifications "are just not present (even though this may not invalidate the final, correct answer)" (Methods). We also find that the mean-score gain from o1 to GPT-5.5 is stable under alternative ways of combining grading votes and weighting scientific fields (Robustness and limitations).

Between the earliest and latest releases, cost per full-mark answer fell from $170.34 to $15.06 (95% CI, $12.81-$18.14), a roughly 11-fold reduction (Figure 2c). This relates expenditure to successful outputs, as in cost-of-pass frameworks[28]. Cost per full-mark answer can be separated into two components: share of answer correct, and cost per answer attempt. The full-mark rate increased 12.6-fold while expenditure per question increased only slightly, from $4.69 to $5.22, so the reduction is driven entirely by higher accuracy, which more than offsets the small rise in expenditure. The lowest cost was $8.19 (95% CI, $6.30-$11.28), achieved by GPT-5.

Token prices and cache discounts also affect which release is cheapest. If we charge cached inputs at the full input price, GPT-5.4 and GPT-5.5 then cost $20.85 and $24.43 per full-mark answer, respectively (Extended Data Figure 6b). The lowest recorded three-phase cost per full-mark answer, US$27.08, is less than 0.3% of the illustrative research salary allocation per publication in the mathematical sciences (US$9,466) and physics (US$11,424), assuming that 20% of working time is spent on research and that two papers are published per year (Methods). These salary allocations cover the production of an entire paper, including formulating a question, deriving a result and verifying it. Our estimate includes only recorded model costs and excludes human labor and independent verification.

AI is also automating tasks in question formulation and verification. Recent systems generate candidate hypotheses at low computational cost[29,30], and some proposals have received experimental support[30,31], although favorable

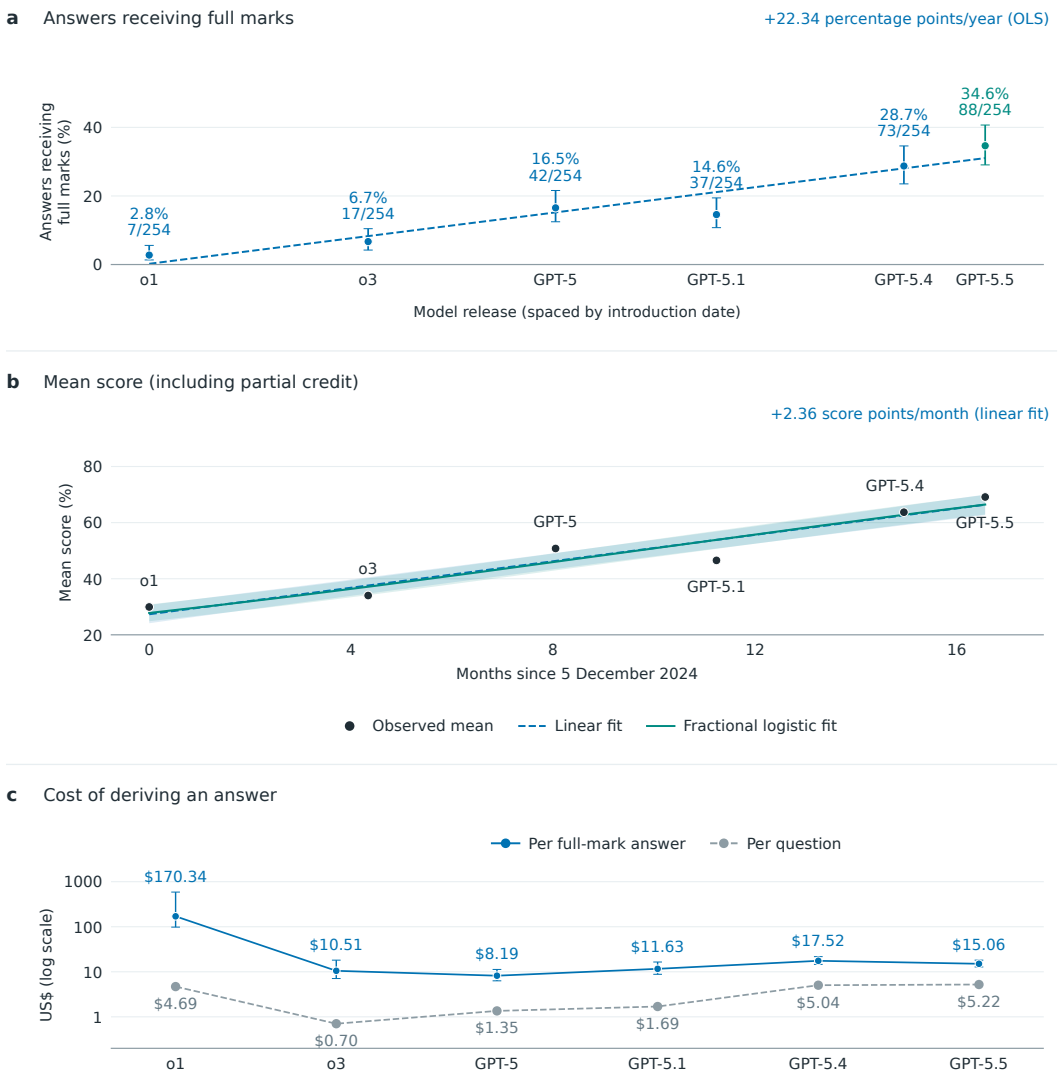


Figure 2: **AI model performance and cost of deriving arXiv paper results.** Each model answers the same 254 questions arXiv-derived questions. **a**, share of AI answers fully replicating human results, with counts and descriptive 95% Wilson intervals. **b**, average scores of AI models on evaluation (allows partial credit). Shading gives pointwise 95% intervals from 20,000 paired bootstrap resamples within fields. **c**, cost to derive AI answers, either per fully correct answer or per attempt. Whiskers give 95% bootstrap intervals for cost per full-mark answer. Costs include unsuccessful attempts and all requested drafts, and exclude human labor and independent verification.

assessments of the proposals do not necessarily predict the quality of the resulting research[32]. AI-assisted formalization is likewise extending formal verification to major mathematical results[33,34], although checking that a proof is correct and checking that it establishes the intended claim remain distinct tasks[18]. Against documented declines in research productivity[35], a sharp fall in the cost of a core step of theoretical research is economically significant in its own right. Our results suggest that, for such questions, language models can now supply candidate derivations at low cost.

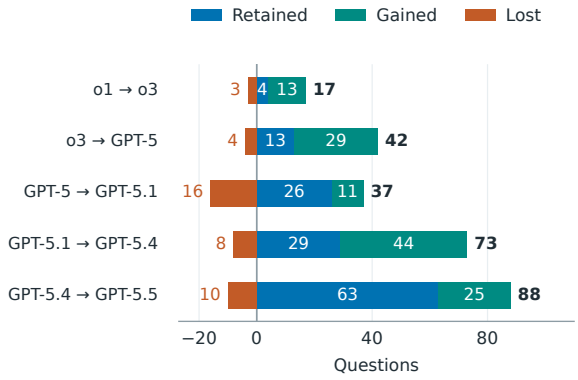
3 Evolution of AI's ability to derive theory papers

The evolution of AI model performance was heavily influenced by changes in extreme outcomes: complete failure or complete success (Figure 3a). The share of questions scoring zero fell from 29.5% with o1 to 5.1% with GPT-5.5, the share receiving full marks rose from 2.8% to 34.6%, and the mean score rose from 29.9% to 69.1%, as demonstrated in Section 2. Partial scores shifted more modestly, with the most common partial score remaining between 50% and 60% for every release. A uniform improvement of about 39 points, the change in the mean, would have moved questions scoring zero into partial credit rather than to full marks. Instead, 20 of the 75 questions on which o1 scored zero reached full marks with GPT-5.5, a rate of 27% compared with 37% for questions o1 answered partially (63 of 172 questions). Questions with partial credit nonetheless remained the main source of progress, accounting for 63 of the 88 full marks obtained by GPT-5.5, although they were also the largest group under o1.

a Distribution of full-and partial-correctness scores (n=254)

	Score (%)										
	Zero (0%)	(0, 10)	[10, 20)	[20, 30)	[30, 40)	[40, 50)	[50, 60)	[60, 70)	[70, 80)	[80, 90)	[90, 100)	Full marks (100%)
o1	**29.5%**	2.8%	7.9%	12.2%	10.2%	5.9%	16.9%	6.3%	2.4%	3.1%	0.0%	**2.8%**
o3	**27.2%**	2.4%	8.3%	10.2%	11.4%	4.3%	17.3%	5.1%	3.1%	2.8%	1.2%	**6.7%**
GPT-5	**13.0%**	2.0%	8.3%	7.9%	9.1%	2.8%	16.9%	5.9%	7.5%	7.9%	2.4%	**16.5%**
GPT-5.1	**17.3%**	0.4%	10.2%	8.3%	8.3%	2.8%	15.7%	9.4%	5.9%	5.5%	1.6%	**14.6%**
GPT-5.4	**6.3%**	1.2%	3.1%	7.5%	8.7%	1.2%	15.0%	7.1%	8.7%	10.6%	2.0%	**28.7%**
GPT-5.5	**5.1%**	0.8%	3.5%	4.7%	6.3%	2.4%	13.8%	5.9%	9.1%	9.4%	4.3%	**34.6%**

b History of whether successive AI models gained or lost the ability to solve additional problems



	Net full-mark answers	Δ cost (US$)	US$ / net extra full-mark answer
o1 → o3	+10	−1,013.69	—
o3 → GPT-5	+25	+165.28	6.61
GPT-5 → GPT-5.1	-5	+86.50	—
GPT-5.1 → GPT-5.4	+36	+848.78	23.58
GPT-5.4 → GPT-5.5	+15	+45.97	**3.06**

Figure 3: **Score distributions and full-mark transitions.** Both panels test 254 questions per model. **a**, percentages receiving zero, partial credit in 10-point bins, or full marks. Color intensity encodes the share of questions; rows sum to 100%. **b**, answers retaining (blue), gaining (green) or losing (orange) full marks relative to the preceding release. Bold totals give the later release's full-mark count. Columns report net gains, changes in total Derivation expenditure and expenditure per net additional full-mark answer when both changes are positive. Dashes mark the remaining cases. Transitions use one recorded outcome per question and release; costs use the documented tariffs (Methods).

This pattern is in line with derivations being chains of steps, in which small gains in reliability per step can compound into large gains in complete answers and an early error can derail the rest of the derivation, although thinking mitigates

this effect[36]. In essence, this means that we are observing thresholding effects, where gradual changes appear slow and then more abrupt[37]. The chain structure of derivations also motivates making numerical and symbolic tools available during *Derivation*, since they are likely a more reliable sub-step than an LLM would be. We find that, on average, o1 makes 0.2 tool calls per answer, whereas GPT 5.5 makes 21.7. That said, we find that using / not using these tools only changes the quality of answers by 3.7 points, which is much smaller than providing more reasoning effort, which increases scores by 13 points (Extended Data Table 3). We compare scores and resource use between configurations with and without tool access to assess this support (Robustness and limitations, Extended Data Table 3).

Progress is also neither monotonic nor cumulative as models evolve over time. Between the first and last releases, there was a net gain in performance, but this masks gains and losses. Two of the seven questions that o1 answers with full marks do not receive full marks from GPT-5.5, and 8 questions that had received some credit fall to zero. GPT-5.1 scores an average of 4.24 points less than GPT-5 on the same questions ($P = 0.023$, paired t-test with Holm adjustment across the 15 release pairs) and has five fewer full-mark answers, although this difference is not significant ($P = 0.44$, exact McNemar test with the same adjustment, Extended Data Figure 3). Its share of zero scores also rises from 13.0% to 17.3% (Figure 3a). Every release fails to reproduce some full-mark answers of the preceding release, and GPT-5.1 is the only one that loses more than it gains (Figure 3b). GPT-5.5 retains 63 full-mark answers from GPT-5.4, gains 25 and loses 10. Of these 25 gains, 20 are on questions that no earlier release had answered with full marks. In total, 107 questions receive full marks from at least one release, 19 of which GPT-5.5 does not answer with full marks. This is the maximum coverage obtainable by selecting among the recorded answers, rather than the measured performance of a routing policy. These gains and losses are consistent with the broader idea of a jagged frontier, in which AI capability varies unevenly across tasks that appear similarly difficult to humans[38]. A new release therefore cannot be assumed to answer every question that its predecessor could. These observed losses may reflect variation between runs or grading error, rather than reproducible declines in capability.

The distinction between complete and partial answers matters for value. When an output requires every component to be correct, its expected value can rise steeply as the reliability of its components improves[39], and our full-mark criterion requires full credit for every rubric requirement and no penalties. We therefore compare changes in total derivation cost with net changes in full-mark answers between consecutive releases (Figure 3b). From o1 to o3, full-mark answers increased by ten while total derivation cost fell by $1,013.69. From GPT-5 to GPT-5.1, full-mark answers decreased by five while cost increased by $86.50. For the remaining transitions, the incremental cost per net additional full-mark answer was $6.61 from o3 to GPT-5, $23.58 from GPT-5.1 to GPT-5.4 and $3.06 from GPT-5.4 to GPT-5.5. These costs depend on both the number of tokens used and the price of those tokens. From GPT-5.1 to GPT-5.4, output nearly doubled to 74.12 million tokens and the output price rose from $10 to $15 per million tokens, which, together with higher input costs, increased total derivation cost from $430.48 to $1,279.25 (Extended Data Table 1; token totals and tariffs in Methods). From GPT-5.4 to GPT-5.5, the output price doubled, but output fell to 34.25 million tokens, reducing output expenditure, and higher input costs left total derivation cost only $45.97 above GPT-5.4 (Extended Data Table 1; Methods). These figures describe the recorded runs at the study's documented costs and do not insinuate that higher expenditure caused the additional full-mark answers.

4 Broad gains across scientific fields

Is this progress shared across scientific fields? Across the seven fields we study, fitted slopes are positive and range from 2.00 to 2.61 score points per month (Figure 4), close to the overall slope of 2.36 (Figure 2b). This implies that AI models could solve roughly an additional quarter of all theory papers in these fields each year. While mean scores differ across fields, their slopes show no statistically detectable difference ($P = 0.83$; Methods). The breadth of these gains suggests that language models could serve as a common input across theoretical research, consistent with the broad applicability associated with general purpose technologies[40]. Our benchmark, however, includes only questions drawn from sampled papers that meet our eligibility criteria (Figure 1b).

Improvements also extend across most of the questions. Between the earliest and latest releases evaluated, scores increased on 207 questions, stayed the same on 27 and decreased on 20, and most questions improved in each field (Supplementary Table S4). High-energy theory is a good example of why levels and gains should be read together. It has the lowest mean with the latest release, 57.9%, but 37 of its 40 questions improved relative to the earliest release, and its mean increased by 42.0 points from 16.0%, the largest increase of all fields (Supplementary Table S4).

Similar mean scores can mask differences in how often a model earns full marks, because the mean retains partial credit. For the latest release, statistics theory and quantum physics have nearly identical means (77.5% and 77.0%), but full marks are more frequent in quantum physics (`quant-ph`), with 18 of 37 questions compared with 14 of 34 in statistics theory (`math.ST`); see Supplementary Table S4. Since the questions and rubrics are held fixed across releases, within-field score changes do not reflect changes in question composition. Differences between fields, in contrast, also

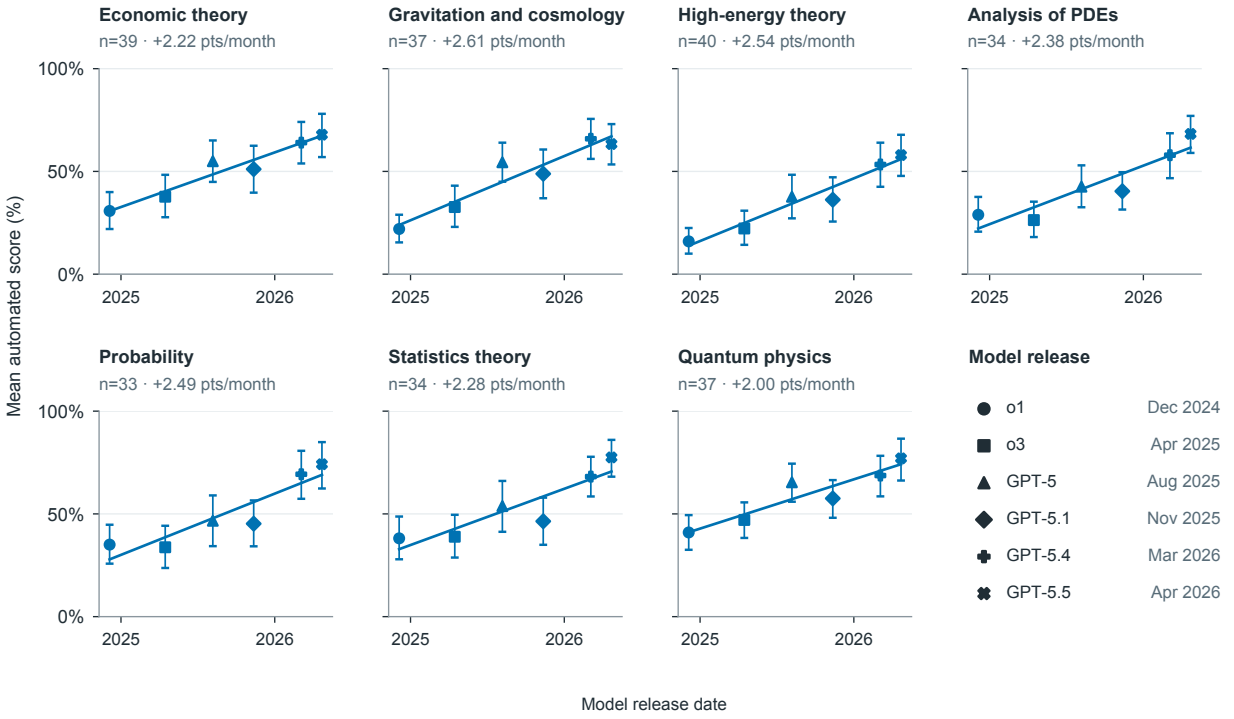


Figure 4: **Mean score trends across scientific fields.** Points show mean scores on the fixed question panel within each field, with model releases identified by shape. Lines show linear fits against first public introduction dates. Labels give sample sizes and slopes in score points per month. Whiskers give pointwise 95% intervals from 20,000 paired bootstrap resamples within fields. The seven fields contain 254 questions in total. PDEs, partial differential equations.

reflect question content, the number of requirements per question, and how close scores already are to 100%, so they describe our selected questions rather than how readily each field could be automated.

5 Robustness and limitations

The mean score gain from o1 to GPT-5.5 is stable across five grading aggregation methods (38.98–39.82 points; Supplementary Table S1) and under equal field weighting or omission of one field at a time (38.68–39.73 points; Extended Data Figure 4a). Aggregation affects individual scores and full-mark classifications (Extended Data Figure 5a-b), and matters more for comparisons between recent releases. GPT-5.5 scores higher than GPT-5.4 under all five methods, but the simultaneous 95% bootstrap interval includes zero when using the median of all votes (-0.26 to 9.05 points), accounting for all 75 contrasts across release pairs and aggregation methods (Methods, Supplementary Table S1).

LLM graders can misjudge reasoning[26] or favor their own outputs[41], and agreement among repeated evaluations does not establish correctness. We use GPT-5.5 as our primary grader and instruct it to scrutinize potential shared errors when evaluating answers generated by the same model (Methods). We have not measured self-preference bias or the effect of this instruction.

Resource and tool use also change across releases. GPT-5.5 scores 18.35 points higher than GPT-5 with similar mean reasoning-token use per question but nearly twice the cumulative input-token use. Compared with GPT-5.4, it shifts from general-purpose code toward theorem checklists and specialized checks (Extended Data Figure 7a-d, Supplementary Note S1). We also compared GPT-5.5 configurations with and without tools on 71 matched questions from analysis of PDEs (`math.AP`) and quantum physics (`quant-ph`). The tool-enabled configuration uses different instructions and scores 3.68 points higher on average (95% CI -1.29 to 8.60; Holm-adjusted $P = 0.15$), with fewer reasoning tokens, more input tokens, and a higher recorded cost (Extended Data Table 3).

Comparisons between releases measure progress in the systems as configured, combining changes in models, inference settings, tools and instructions without isolating their contributions (Methods, Extended Data Table 1). To limit contamination, we use source papers first posted after the latest documented knowledge cutoff among the evaluated models (Methods).

Our cost estimates measure recorded expenditure on deriving known results, including failed attempts. Grading decisions determine which answers count as full-mark outputs and therefore affect both cost per full-mark answer and incremental cost per additional full-mark answer. Human labor, including the work needed to verify and use the outputs, is excluded.

6 Research allocation under changing costs

We find that models receive full marks on a growing share of well-posed questions from recent theoretical research at low recorded derivation cost. Scientists must also formulate questions, gather evidence, verify conclusions and place results in context. Holding the organization of research fixed, aggregate productivity gains depend on the economic shares of the affected tasks and the savings achieved within them[42]. Applying this framework requires accounting for the work needed to verify and use AI outputs.

Lower costs could also make more research worthwhile, increasing demand for complementary expertise. Employment effects depend on both labor savings and the expansion of research, alongside scientists' teaching, mentoring and administrative responsibilities. More broadly, AI may provide a general-purpose method of invention and reshape the organization of R&D[43]. We consider three settings in which lower derivation costs could change the scale and direction of research.

First, there may be an overhang of important questions that models could answer cheaply once formulated. The opportunity would be to identify these questions and connect their answers into research programs. Mathematics may offer such opportunities because many discoveries do not require new observations about the physical world. Cheaper intermediate results could also open previously costly paths toward larger scientific goals. Just as economic incentives shape the direction of innovation[44], lower derivation costs could change which questions are pursued and the amount of research conducted along existing paths.

Second, progress may be limited by the evidence we can collect. Observations needed to distinguish competing explanations may be unavailable, while experiments that test conclusions or reveal new questions may remain costly. Aghion, Jones and Jones show how essential tasks that remain difficult to automate can constrain the benefits of automation when tasks are sufficiently complementary[45]. In scientific research, costly experiments or inaccessible observations could impose a similar constraint.

Better cognition may nevertheless help scientists design more informative experiments or learn more from available observations. Agrawal, McHale and Oettl model discovery as a prioritized search in which tests are costly. Better predictions can increase the probability of discovery and shorten search, although realizing these gains may require investment in testing capacity[46]. The same authors also show that better predictions can increase or decrease downstream testing, depending on the innovation problem[47]. Cheaper cognition could therefore make some experimental research programs viable even as better predictions reduce the testing required within others.

Third, cognition may remain a constraint even when evidence is available and verification is inexpensive. Our work does not establish how reliably models can generate productive ideas, create useful representations, sustain a line of inquiry or solve the hardest scientific problems. Delegating more tractable derivations to models could allow scientists to devote more time to those tasks.

Scientific progress also depends on which results the community has time to assess and build on. In mathematics, a declaration initially signed by 25 Fields Medalists warned that rushed announcements of AI-generated solutions leave insufficient time to carefully write up results, to isolate new methods and ideas and acknowledge prior work[48]. Evidence from one journal links AI-associated writing to increasing submissions and greater demands on editorial assessment[49], while submission limits at arXiv reflect pressure on moderation capacity[50]. If production grows faster than assessment, a larger literature need not yield proportionately more usable knowledge.

If derivation costs continue to fall faster than verification costs, choosing questions and establishing reliable answers may become stronger constraints on theoretical research. We provide a reference point for tracking derivation costs with publicly available models. Comparable measurements of verification costs would help assess how the balance between producing and establishing results changes over time.

In summary, we find rapidly improving AI capabilities for completing theoretical work across 7 fields of science. In each of these fields, we find that performance is improving by roughly 2 percentage points per month, meaning that we

might expect AI's ability to fully solve these problems to rise from 35% with GPT 5.5 to 60% a year from now. And because these derivations are relatively cheap ($15), we expect rapid adoption of these techniques. Such rapid changes are likely to cause some disruption to established norms and ways of working in these fields, resulting in changes to the tasks that theorists spend their time on and changes to how theory research is done.

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Methods

Sampling frame and panel assembly

We sampled from first-version arXiv papers posted between 1 December 2025 and 16 May 2026 across the seven primary categories math.AP, math.PR, math.ST, econ.TH, hep-th, gr-qc and quant-ph. We used a static, affiliation-filtered spreadsheet of potential papers. The filter used model-generated lists of 100 institutions in each subfield, and kept papers that had an author with an affiliation in these lists. These lists were a curation tool, not a bibliometric ranking. Sampling was done within this frame, not uniformly from all arXiv papers or all research in these areas.

Our pool generator grouped the spreadsheet by primary category, sampled pseudorandomly with recorded seeds, and removed already sampled identifiers from the pool. The archived pool generator uses 35 as the target number of papers to accept per category, and 30 as the minimum, rather than fixed final field sizes. Round 1 assessed 35 papers per category. Round 2 added 10, 19, 35 and 35 papers to assess in econ.TH, hep-th, gr-qc and quant-ph, respectively; inclusion counts are in Supplementary Table S2.

We selected one answer and one grading result for each question and model, resulting in 1,524 primary observations. We did not impute any scores. The fixed panel has first-posting dates ranging from 3 December 2025 to 14 May 2026, which are all after the latest documented model knowledge cutoff, 1 December 2025, following the approach used by LiveBench[51]. Candidate metadata, assessment decisions and panel membership are provided in Supplementary Data 1.

Curation and information separation

The hidden grading material included what was required in the answer, information to reference, instructions for grading, and explanations of how to arrive at the correct answer. Where the supplied assumptions came from, and why they were given, are included in the hidden grading material. We used validation for structured output, and audit and repair steps. The released configuration and stage records identify which steps ran for each item. Generating this package once per paper does not mean one call was made to Curation, because there could be multiple calls to extract, generate, audit and repair.

Evaluation does have access to paper metadata, such as identifiers, when grading, although it does not receive the manuscript. Therefore, we do not fully blind paper identity. The public inputs and prompts that we released show what information is given to each stage. Prompt templates and message-assembly code are indexed in the Supplementary Information; complete question packages are in Supplementary Data 1.

Model configurations and draft selection

We compare o1, o3, GPT-5, GPT-5.1, GPT-5.4 and GPT-5.5, with reasoning effort set to `high` for the first four and `xhigh` for the last two. These effort levels do not correspond to the same amount of computation. We asked each model to generate three drafts, each with a maximum of 15 turns for the main agent and 50 tool calls across its stages. Conditional critique, revision and repair stages had separate turn limits, specified in the recorded configurations.

Knowledge cutoffs and available snapshots with dates are taken from the model documentation[52]. The aliases that we recorded do not guarantee which snapshot was used for each call. We report when models were introduced, when they became available on the API, and when artifacts were saved, separately in Extended Data Table 1.

We select one of the three drafts, without combining information from the other drafts or using the grading scores. The default ranking is lexicographic, where we prefer nonempty answers without execution errors, then designated empty gap outcomes, and then execution failures. We then prefer more lines of verification, greater coverage of detected verification triggers, fewer markers of possibly invented entities, fewer structured gaps, fewer compact gaps, and lower latency. Trigger coverage is the number of verification lines divided by detected triggers, capped at one and set to zero when no trigger is detected. If all of these are tied, we pick the first draft generated. Hidden rubric content does not enter the ranking.

Tool and resource records

The theorem-hypothesis checklist is not a proof assistant. It uses the model being evaluated to check if a theorem's assumptions are met, alongside deterministic checks for specified cases. Some other tools also use the model to help with verification, including encoding claims for numerical, symbolic or solver-based checks. The tool implementations are included in Supplementary Code 1. Code Interpreter is not available for o1. Recorded configurations determine the available tools and instruction overlays.

We count tool usage by looking at the number of calls that were completed, and report the percentage of questions that had a tool call and the number of tool calls per question, including questions with no tool calls (Extended Data Figure 7c-d, Extended Data Table 2). Counts include linked Derivation runs and calls returning errors or inconclusive results. For each tool, we count a question once when reporting question-level use, but include repeated calls in invocation totals. We sum provider-reported token usage across all linked calls, so cumulative input includes instructions, earlier messages and tool results presented again on subsequent calls.

Evaluation and scoring

We use GPT-5.5 at `xhigh` to grade all of the selected answers. If the model that answered the question is the same as the model that is grading, we add a prompt that tells it to look for unsupported standard steps, algebra and mechanisms that seem plausible, using the answer key. The configured Derivation and Evaluation model names are compared after trimming whitespace. The instruction and its dispatch condition are included in the Supplementary Information prompt index.

We grade each answer with three evaluations, and if they do not agree on a requirement's status, we grade it two more times and combine the results. We have 986 answers that were graded three times and 538 that were graded five times. If a grading output fails JSON or schema validation, we retry it, allowing up to three attempts per vote including the initial attempt, and use the outputs that are correctly parsed.

We align elements by position when their nonempty lists have equal lengths; otherwise, we match them by their normalized descriptions. In the latter case, if at least half of the evaluations list an element, we count missing entries in other evaluations as not met; otherwise, we use only the evaluations that list it. We combine these by taking the majority vote, and if there is a tie, we mark the element as not met.

We also have a rule that can decrease the score of a requirement. If at least 30% of the evaluations assign a lower requirement status, we take the lowest of those statuses. The highest score for each question is 10, and each requirement is weighted equally. If a requirement is fully, partially or not met, we give it full, half or no points, respectively.

We subtract from this total if the answer has any invalidity or scope issues. Invalidity is when the model makes a false claim that is not covered by the requirements, and we tell the grader not to penalize the same error twice. Scope applies if the model gives a tangential answer, does not answer the question asked, or refuses to answer. We can subtract up to half of the maximum score for each. We take the median across evaluations for each penalty, and the most frequent category for scope, with ties resolved toward the stricter option. If the answer is in scope, there is no scope penalty. If no validity errors are recorded across the evaluations, there is no validity penalty. We recalculate the score, cap it between 0 and 10, and then scale it to 0–100.

Human review

To assess how faithful the AI judging system is, we created an invite-only web platform for expert review of graded answers. The platform shows to the researcher only the answers the automated AI grader gave the maximum score. Partially correct and incorrect answers are not uploaded. The human reviewer can therefore detect full-mark answers that do not deserve maximum score. A reviewer signs in with name and e-mail, academic institution and the MSC 2020 areas they are comfortable grading, and receives a few arXiv papers. For example, for analysis of PDEs (math.AP) the queue contained the eight GPT-5.5 answers that received full marks.

Each item page in the platform shows, in a fixed order, (i) the public question with its assumptions, exactly as given to the model, (ii) the recorded answer with the full derivation, (iii) the verification tools the model ran, with their outputs, (iv) the hidden rubric with its answer keys and required derivation steps, and finally (v) the grader's verdict. The reviewer therefore sees the automated grade and is not blinded to it.

The reviewer annotates five surfaces. For the question, whether it is well-posed, faithfully extracted and answerable without the paper. For the derivation, whether it is correct and complete. For the tool checks, whether they were applied correctly to the load-bearing claims. For the overall verdict, whether the answer earns the recorded score. For each answer key, whether the reviewer agrees with the grader's verdict on that key. The platform allows confirm, partial or dispute assessments, accompanied by free-text comments. Annotations save automatically and the reviewer marks each item complete explicitly.

One specialist in the analysis of PDEs reviewed five papers and considered all five answers correct or apparently correct. He agreed with the full-mark verdict on four answers; for the fifth, he indicated that he might give a lower score because some justifications were missing, while noting that this might not invalidate the correct final answer. The question was confirmed as well-posed and faithfully extracted on all five items; two comments note that the extracted question

uses unconventional vocabulary. The derivation comments also ask for clearer explanations of some steps, as a referee would. Table 1 summarizes the specialist's assessment of correctness and agreement with the automated score.

Item	Assessment of the answer	Agreement with full marks
2601.12653	Correct	Agrees
2601.16482	Correct	Agrees
2603.21367	Appears correct	Agrees
2603.28194	Correct	Agrees
2604.01895	Appears correct	Would consider a deduction

Table 1: Specialist review of five GPT-5.5 full-mark answers in analysis of PDEs. The specialist considered all five answers correct or apparently correct and agreed with full marks on four. For item 2604.01895, he indicated that missing justifications could warrant a lower score without necessarily invalidating the correct final answer. Assessments summarize the reviewer's comments and distinguish correctness from agreement with the automated score.

Expenditure accounting

We calculate costs using the usage reported by the provider and our archived cost table. We separate uncached input, cached input and output tokens, and multiply each by its archived tariff. We report reasoning tokens as part of output tokens, without counting them twice. We can link calls to a stage, model and question using the response identifier that the provider gives us. If a response is shared, we only count it once within each accounting boundary. We include all of the work that Derivation did, including linked preceding runs, failed answers and calls that led to failed or repeated attempts, not just the selected draft. Evaluation costs also include retries.

We add the cost of Evaluation in a separate scenario. We also add the cost of Curation, US$737.25, to each model, assuming we are only using one model. We count this shared cost once when aggregating the study's expenditure. For incremental cost, we divide the change in total Derivation expenditure by the net change in full-mark answers between consecutive releases, and report the ratio only when both changes are positive.

We have a scenario that charges cached input at the same rate as uncached input. We also have scenarios that use the same cost for input, and vary the ratio of output to input and of cached to uncached input. We use a base cost of US$1 per million input tokens, output-to-input ratios of 1, 2, 4, 8 and 16, and cached-to-uncached ratios of 0, 0.1, 0.5 and 1, for a total of 20 scenarios.

Salary allocation

We use the annual mean wage from the BLS, for May 2025, in the Colleges, Universities, and Professional Schools industry, to calculate a salary allocation per publication. Postsecondary mathematical science teachers have a mean wage of US$94,660, and postsecondary physics teachers have a mean wage of US$114,240[53,54]. We multiply each wage by the assumed research-time share of 20% and divide by two papers per year. Benefits are excluded.

Resampling and paired comparisons

We resample by question, unless otherwise stated. We take 20,000 bootstrap samples within each field, keeping the number of questions per field and the results of all models for each question. Except for the simultaneous intervals described below, we report 95% confidence intervals using the 2.5th and 97.5th percentiles. We use Wilson intervals[55] for our main full-mark and inclusion percentages. We bootstrap the cost ratios by resampling the cost and result of each question together. If a resample has no full-mark answers, we report an infinite ratio instead of dropping the sample. Random seeds are specified in the released analysis scripts.

We use McNemar's test, with exact P values, to compare full marks, and a paired t-test to compare mean scores; both are two-sided. We also use a Wilcoxon signed-rank test, with the Pratt treatment of zeros, as a sensitivity analysis. We adjust for multiple comparisons using Holm[56] across the 15 comparisons between releases for each outcome. We have three more comparisons between configurations—GPT-5.5 with versus without tools, GPT-5.5 at xhigh versus low effort, and GPT-5.4 versus mini—which we adjust separately across the three tests for each outcome, using only the scored questions that they have in common. We calculate token ratios by dividing condition means and recompute them in the paired bootstrap.

Sensitivity to grading aggregation

For our grading sensitivity, we compare our rubric aggregation rule to the mean and median of either the first three evaluations or all evaluations, using the recorded per-vote scores. We have 15 comparisons between releases for each of the five methods, for a total of 75. We take 100,000 bootstrap samples, pairing the releases and stratifying by field. For each contrast, we estimate its bootstrap standard error as the standard deviation across resamples. We center each resampled contrast on its observed value and divide by this standard error. We take the 95th percentile of the maximum absolute standardized contrast across all 75 comparisons as a common critical value. Each simultaneous 95% interval is the observed contrast plus or minus this critical value times its bootstrap standard error. We repeat the analysis with different random seeds to assess sensitivity to Monte Carlo variation.

We use Holm separately for each of the paired t, sign, Wilcoxon, and exact McNemar tests across the 75 comparisons. The Wilcoxon tests in this analysis omit zero differences. We recalculate full-mark classifications under each aggregation method.

Release-date trends and field comparisons

We fit a linear and a fractional logistic model to the release dates and the mean score of each model. We use first public introduction dates, measured from 5 December 2024, and express monthly slopes using $365.25/12$ days per month. We fit the linear model by ordinary least squares and the fractional logistic model by Bernoulli quasi-maximum likelihood on scores scaled to $[0, 1]$[57]. We refit both models in each paired question bootstrap to obtain pointwise confidence intervals. We test for field heterogeneity by taking the slope of each question across the six releases and fitting a regression with field as a predictor. We use a joint Wald test with HC3 to test whether the mean slopes differ across fields.

We compare field compositions by weighting fields equally or omitting one field at a time, keeping question-level results paired. We count answers scoring at least 75%, 90%, 95% or 100% for the threshold analysis. We also look at what happens if we remove the questions that were posted in December 2025, in an exploratory analysis (Supplementary Table S3). We retain existing answers and scores and resample within the remaining field counts. We apply Holm adjustment across the two endpoints, mean score and full-mark rate, for this exclusion; the all-question reference retains the primary adjustment across 15 release pairs.

AI-assisted preparation

Codex (OpenAI) assisted with analysis code, numerical checks, and manuscript revision.

Data availability

The 254 question packages, selected model answers, grading records, scores and cost data are provided in Supplementary Data 1, together with the numerical source data for the figures and tables.

Code availability

The analysis and figure-generation scripts, pipeline source, prompt templates and recorded configurations are provided in Supplementary Code 1. The Supplementary Information includes an index of the prompts and message-assembly code.

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Author contributions

G.M. and N.T. conceived the study, interpreted the results, and supervised the project. G.M. designed the benchmark and evaluation protocol, developed the system and prompts, conducted the model evaluations, performed the statistical analyses, and prepared the figures. C.M.V. developed the independent review system. G.M., N.T. and C.M.V. wrote the manuscript and reviewed it.

Competing interests

The study received support from OpenAI, whose models are evaluated in this work, as described in the Funding statement.

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Extended Data Figure 1: **Coverage, penalties and zero scores.** All panels use 254 questions per release. **a**, Mean coverage before penalties and final score. Error bars give 95% intervals for the final score from 20,000 paired bootstrap resamples within fields. **b**, Mean validity and scope deductions in score points. **c**, Shares of all questions scoring zero, separated into answers with no coverage and answers whose positive coverage is cancelled by penalties; counts appear on the right. Scores are floored at zero, so subtracting mean deductions from mean coverage need not recover the mean final score.

Rao-Blackwellized e-variables

arXiv:2512.16759 [math.ST] Abridged frozen package

Public input

Available to Derivation

Question How does conditioning an e-variable on a sufficient statistic compare with the original e-variable in terms of e-validity and expected concave utility?

RA-1 There is a family of probability measures $\{P_\theta\}_{\theta\in\Theta}$ on a measurable space $(\Omega, \mathcal{F})$, with a designated null subset $\Theta_0 \subseteq \Theta$; an e-variable for Θ_0 is a nonnegative extended random variable E satisfying $\mathbb{E}_\theta[E] \leq 1$ for all $\theta \in \Theta_0$.

RA-2 The statistic S is sufficient for the full family $\{P_\theta : \theta \in \Theta\}$ in the sense that conditional expectations of measurable functions given S admit a version independent of θ whenever those conditional expectations exist.

RA-3 The conditioned version under study is $G = \mathbb{E}_\Theta[E \mid S]$; concave utility functions $f : (0, \infty) \to \mathbb{R}$ are extended to $[0, \infty]$ by endpoint limits, and utility comparisons are made only when the two expected utilities are well-defined extended values.

Hidden grading material

Withheld from Derivation

Answer keys

Two keys; 5 points each

AK-1 The conditioned variable $G = \mathbb{E}_\Theta[E \mid S]$ is e-valid for the null: it is nonnegative and satisfies $\mathbb{E}_\theta[G] \leq 1$ for every $\theta \in \Theta_0$.

AK-2 For every $\theta \in \Theta$ and every concave utility f in the stated endpoint convention, if $\mathbb{E}_\theta[f(E)]$ and $\mathbb{E}_\theta[f(G)]$ are well-defined, then $\mathbb{E}_\theta[f(G)] \geq \mathbb{E}_\theta[f(E)]$.

Accepted derivation

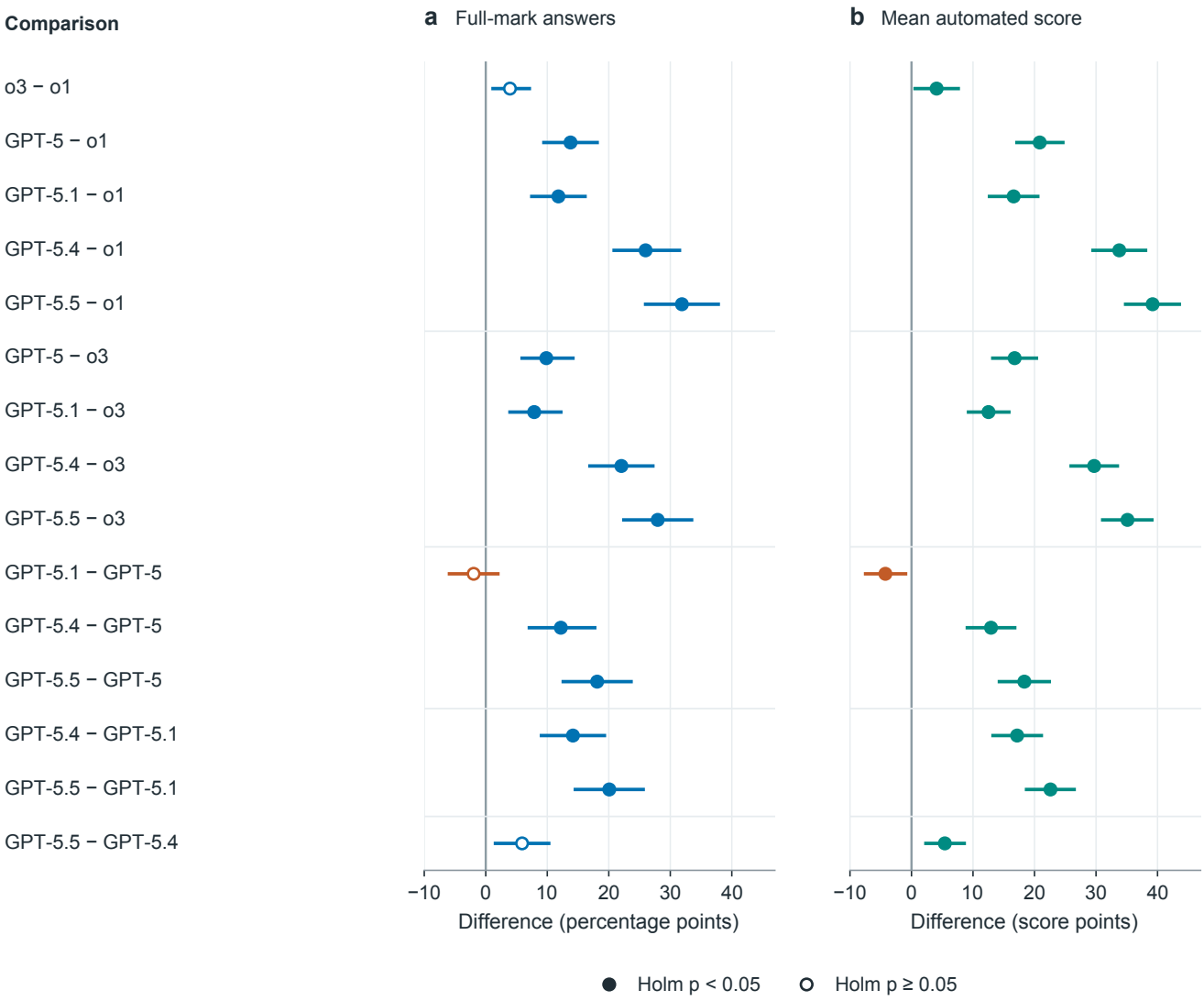
The authors set $G = \mathbb{E}_\Theta[E \mid S]$. Sufficiency lets them use the same conditional expectation under every P_θ. The tower property gives $\mathbb{E}_\theta[G] = \mathbb{E}_\theta[E]$, yielding e-validity under the null, and conditional Jensen gives $\mathbb{E}_\theta[f(E)] \leq \mathbb{E}_\theta[f(G)]$. Lemma A.1 supplies the extended-valued Jensen statement for nonnegative, possibly nonintegrable variables.

Diagnostic context: strictness (FM-SI)

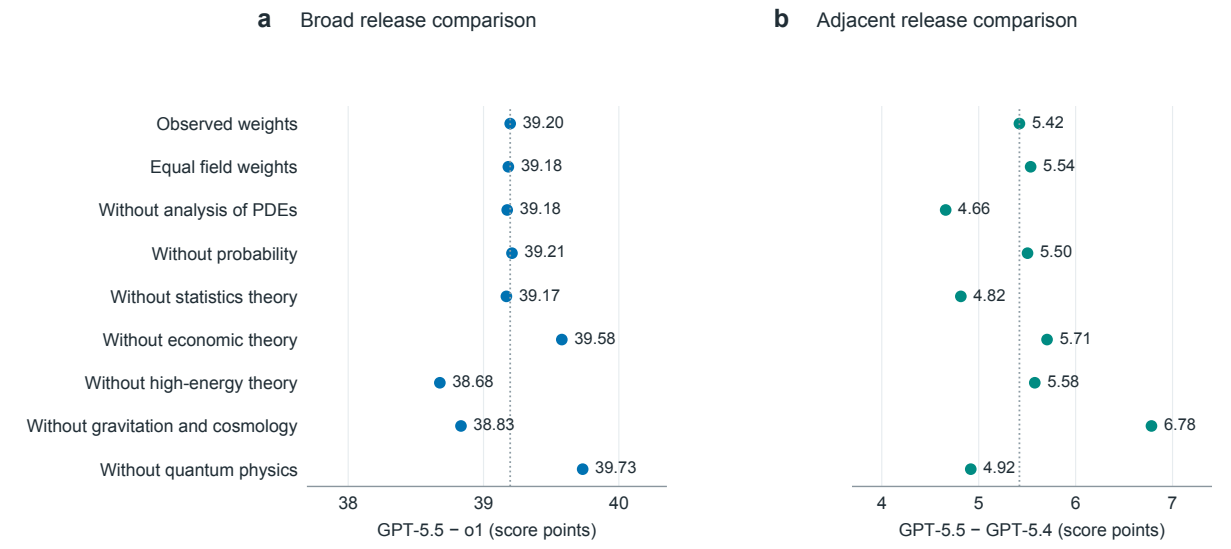
The theorem guarantees non-decrease, not strict increase; equality can occur, for example when E is already S-measurable or when the utility is affine on the relevant range.

Question, assumptions and both key commitments are reproduced verbatim.
Selected hidden context is also verbatim; other rubric fields are omitted.

Extended Data Figure 2: **Public input and hidden grading material for a statistics question.** Abridgment of the frozen package for arXiv:2512.16759. Derivation receives the question and three premises; Evaluation additionally receives two equally weighted answer keys. The keys require e-validity by the tower property and concave-utility dominance by conditional Jensen, without requiring strict improvement. The premises specify endpoint conventions and when the relevant expectations are defined. The complete package is provided in Supplementary Data 1.



Extended Data Figure 3: **Paired differences between model releases.** All 15 newer-minus-older comparisons use the same 254 questions. **a**, Differences in full-mark rates, in percentage points. **b**, Differences in mean scores, in score points. Bars give marginal 95% intervals from 20,000 paired bootstrap resamples within fields. Filled markers indicate Holm-adjusted $P < 0.05$, using two-sided exact McNemar tests in **a** and paired t tests in **b**, with adjustment across 15 comparisons separately for each outcome. Orange marks negative differences. Intervals are not adjusted for multiple comparisons.



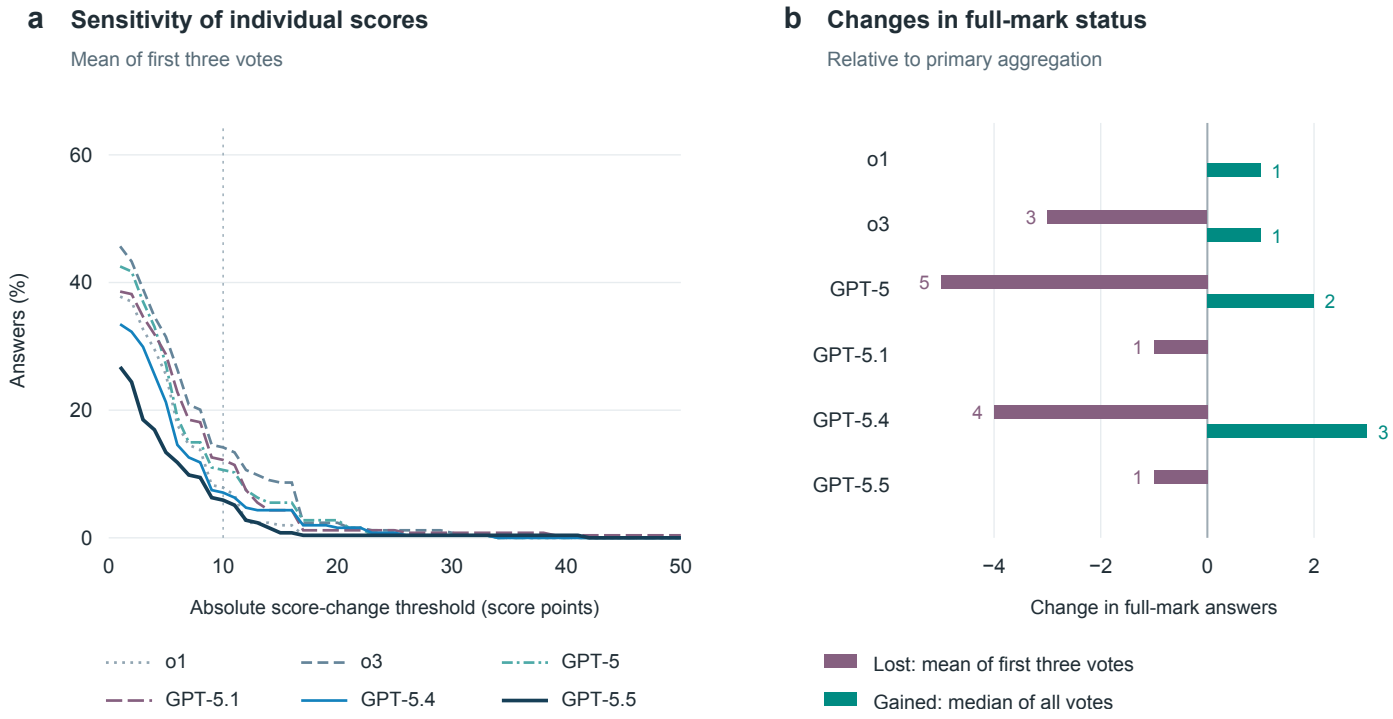
c Answers meeting alternative score thresholds

Counts out of 254

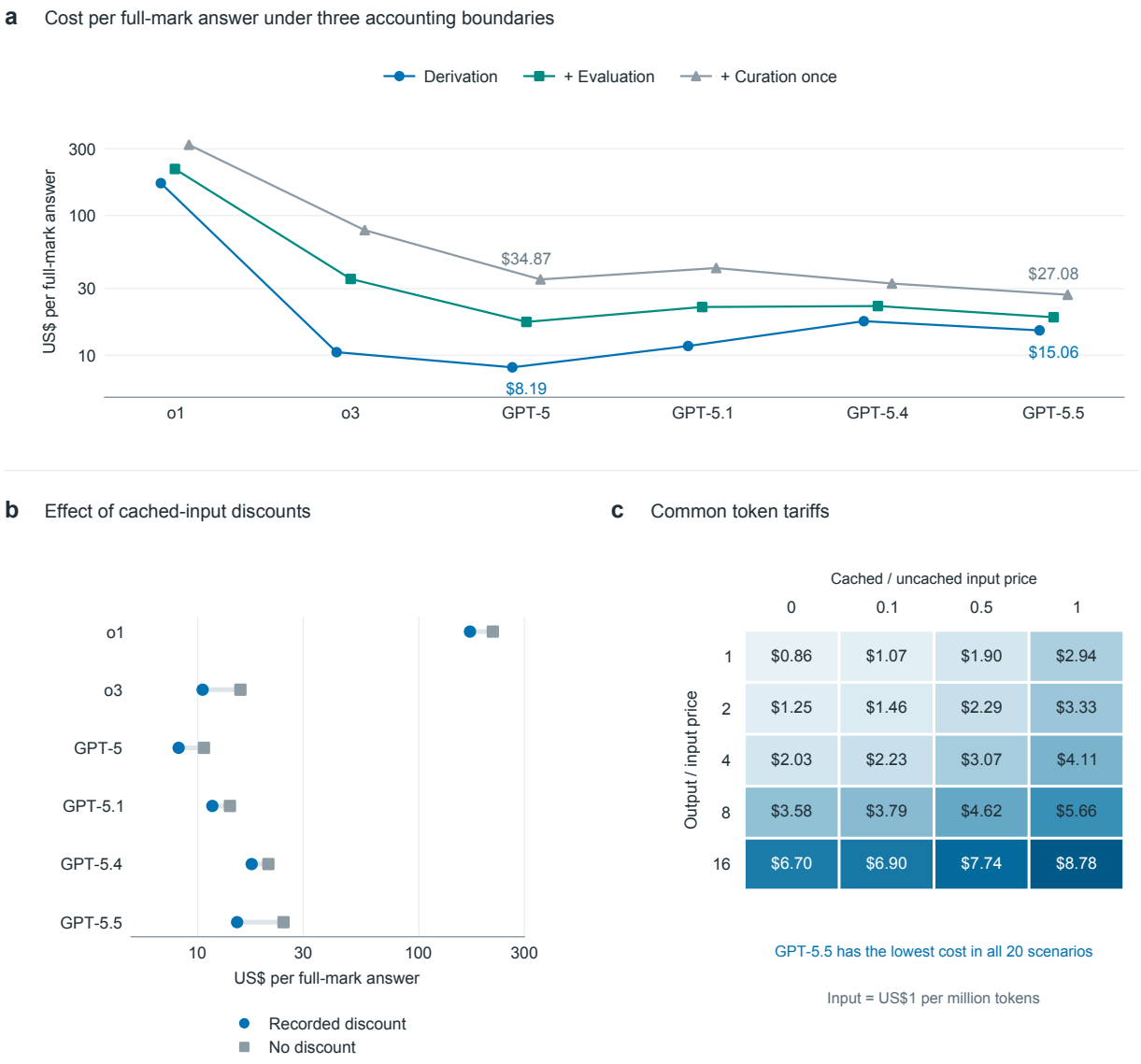
	≥75%	≥90%	≥95%	100%
o1	19	7	7	7
o3	31	20	17	17
GPT-5	85	48	42	42
GPT-5.1	66	41	37	37
GPT-5.4	118	78	73	73
GPT-5.5	142	99	88	88

Lower thresholds include partial-credit answers.

Extended Data Figure 4: **Sensitivity to field composition and score thresholds. a,b**, Mean-score differences for GPT-5.5 versus o1 (**a**) and GPT-5.5 versus GPT-5.4 (**b**), using observed field weights, equal field weights or omission of one field at a time. Points are paired differences under each weighting scheme. **c**, Counts meeting score thresholds of 75%, 90%, 95% and 100%, out of the same 254 questions per release.



Extended Data Figure 5: **Sensitivity to grading aggregation.** Both panels use 254 answers per release. **a**, Percentage of answers whose absolute score change from the primary aggregation reaches the indicated threshold when using the mean of the first three votes. The dotted line marks 10 score points. **b**, Full-mark classifications lost under the mean of the first three votes (purple) and gained under the median of all votes (green), relative to the primary aggregation.



Extended Data Figure 6: **Cost rankings under alternative accounting and token prices.** All panels hold the 254-question panel and recorded answers fixed. **a**, Cost per full-mark answer for Derivation alone, Derivation plus Evaluation, and all three stages. The full shared Curation cost is assigned once to each hypothetical single-model deployment. **b**, Derivation cost per full-mark answer with recorded cache discounts and with all input charged at the uncached rate. **c**, Lowest Derivation cost per full-mark answer under 20 common-tariff scenarios, with uncached input priced at US$1 per million tokens. GPT-5.5 is the lowest-cost model in every scenario; cell values give its cost. Token use is unchanged across pricing scenarios.



Extended Data Figure 7: **Recorded token and tool use across model releases.** All panels use 254 questions per release. **a**, Mean cumulative reasoning tokens per question. **b**, Mean cumulative input tokens, split into cached and uncached input. Error bars in **a,b** give 95% intervals for reasoning and total input tokens, respectively, from 20,000 paired bootstrap resamples within fields. **c**, Percentage of questions with a completed invocation of each tool. **d**, Mean completed calls per question, including questions with no calls; totals appear below. The final row groups 11 tools, counting questions using any of them in **c** and summing their calls in **d**. N/A marks unavailable tools. Completed calls may return errors or inconclusive results.

Extended Data Table 1: **Model configurations, dates and reconstructed token costs.**

Configuration	Effort	Tools	Scored n	Derivation (US$)	Evaluation (US$)
o1	high	Yes	254	1,192.40	314.08
o3	high	Yes	254	178.70	419.08
GPT-5	high	Yes	254	343.98	383.19
GPT-5.1	high	Yes	254	430.48	388.47
GPT-5.4	xhigh	Yes	254	1,279.25	362.18
GPT-5.5	xhigh	Yes	254	1,325.22	320.33
GPT-5.4-mini	xhigh	Yes	253	441.40	295.12
GPT-5.5, no tools	xhigh	No	71	347.00	86.37
GPT-5.5, low	low	Yes	34	65.97	43.51

Release	Introduction	API availability	Documented snapshot
o1	2024-12-05	2024-12-17	o1-2024-12-17
o3	2025-04-16	2025-04-16	o3-2025-04-16
GPT-5	2025-08-07	2025-08-07	gpt-5-2025-08-07
GPT-5.1	2025-11-12	2025-11-13	gpt-5.1-2025-11-13
GPT-5.4	2026-03-05	2026-03-05	gpt-5.4-2026-03-05
GPT-5.5	2026-04-23	2026-04-24	gpt-5.5-2026-04-23
GPT-5.4-mini	2026-03-17	2026-03-17	gpt-5.4-mini-2026-03-17

Configuration	Earliest saved artifact	Latest saved artifact
o1	2026-06-10	2026-09-04
o3	2026-06-10	2026-06-11
GPT-5	2026-09-04	2026-09-04
GPT-5.1	2026-08-28	2026-09-04
GPT-5.4	2026-09-05	2026-09-05
GPT-5.5	2026-06-06	2026-06-10
GPT-5.4-mini	2026-06-09	2026-06-10
GPT-5.5, no tools	2026-06-09	2026-06-10
GPT-5.5, low	2026-06-10	2026-06-10

The six primary configurations provide 1,524 scored answers across seven fields; all configurations provide 1,882 scored answers across 52 configuration–field datasets. GPT-5.4-mini has one additional unscored record whose costs are included. The GPT-5.5 tool and effort comparisons cover 71 questions in two fields and 34 questions in one field, respectively. Code Interpreter is unavailable to o1. GPT-5.5 at xhigh performs Evaluation throughout. Saved-artifact dates identify files containing the selected responses, rather than complete call timestamps. Costs include linked preceding Derivation runs and Evaluation retries, with reused grading responses counted once. Shared Curation costs US$737.25 once overall and is excluded from these columns, along with human labor, hosted execution charges and unrecorded computation. Documented snapshots are not authenticated runtime identities. Token totals and tariffs are supplied in the files identified in Methods; configuration and prompt details are provided in the Supplementary Information prompt index.

Extended Data Table 2: **Recorded use of the 18 invoked tools.**

Tool	o1	o3	GPT-5	GPT-5.1	GPT-5.4	GPT-5.5	Calls
Theorem checklist	29	184	210	168	157	210	5,864
Code Interpreter	—	80	29	64	126	62	5,376
Counterexample search	1	48	16	9	131	189	1,760
Scaling limits	1	17	19	9	76	137	1,383
Series expansion	0	24	26	28	49	48	704
Negative-claim audit	6	19	11	11	51	106	609
Dimensions	0	14	4	0	15	41	201
Tensors and curvature	0	1	1	0	2	9	33
Linear programming	0	1	1	0	2	0	13
Mechanism conditions	0	1	0	0	1	1	10
Mechanism encoding	0	1	0	0	1	1	10
Field covariance	0	4	0	0	1	2	9
Large deviations	0	0	0	0	0	1	9
Linear operators	0	3	0	0	2	1	6
Markov chains	0	0	0	0	0	1	2
Energy conditions	0	1	0	0	0	0	1
PDE classification	0	0	0	0	0	1	1
Quantum-channel encoding	0	1	0	0	0	0	1

Model columns count questions with at least one completed invocation out of 254; a question may use several tools. The final column sums calls across all six releases, including repeated invocations and linked preceding runs. A dash marks an unavailable tool; zero indicates no recorded use. Completed calls may return errors or inconclusive results.

Extended Data Table 3: **Score and resource differences between paired configurations.**

Comparison	n	Mean score difference (points)	Reasoning tokens ratio	Input tokens ratio	Adjusted P
Tools on versus off		$+3.68$	0.88	5.45	
PDEs and quantum physics	71	$[-1.29, 8.60]$	$[0.83, 0.92]$	$[5.02, 5.90]$	0.153
Xhigh versus low effort		$+12.98$	20.51	2.13	
PDEs	34	$[2.48, 23.83]$	$[18.41, 22.88]$	$[1.93, 2.35]$	0.0527
GPT-5.4 versus mini		$+19.22$			
All seven fields	253	$[15.07, 23.35]$	—	—	3.82×10^{-16}

Brackets give marginal 95% intervals from 20,000 paired bootstrap resamples within fields. Score differences subtract the second condition from the first; token ratios divide their means. P values come from two-sided paired t tests, with Holm adjustment across the three score comparisons. The first two comparisons use GPT-5.5. Tools on versus off holds effort at xhigh but changes instructions; xhigh versus low holds tools available. Mean Derivation costs per question are US$5.18 versus US$4.89, US$5.77 versus US$1.94, and US$5.04 versus US$1.74, respectively. The unscored mini record is excluded from the paired comparison. Dashes mark unreported ratios.

Supplementary information

AI Disruption and Falling Costs in Theory Research

This supplement contains two supplementary notes, four supplementary tables and an index of the accompanying prompt files and message-assembly code.

Supplementary Note S1: token use and tool choices

Mean recorded reasoning-token use per question rises from about 17,000 for o1 to 90,000 for GPT-5, stays near 87,000 for GPT-5.1, peaks at 249,000 for GPT-5.4 and falls to 92,000 for GPT-5.5 (Extended Data Figure 7a). Cumulative input tokens sum the input across all Derivation calls, not only the public input (Extended Data Figure 7b). They grow because successive calls repeat instructions, previous messages and tool results, and revision calls also receive the generated draft. From GPT-5.4 to GPT-5.5, the increase in mean cumulative input per question comes from cached input, which the provider reports as reused from earlier requests, while uncached input falls from about 221,000 to 162,000 tokens, a decrease of 26.6%.

Completed tool calls, which include calls returning errors or inconclusive results, average 6.5–6.9 per question for o3, GPT-5 and GPT-5.1, compared with 20.8 for GPT-5.4 and 21.7 for GPT-5.5 (Extended Data Figure 7c,d). The earliest release, o1, makes little use of its available tools. Between GPT-5.4 and GPT-5.5, total call counts are similar but tool choice changes. Use of the Python execution tool falls from 49.6% to 24.4% of questions, and its mean calls per question from 10.34 to 2.16. Use of the theorem-hypothesis checklist, which combines model-based review with deterministic checks for specific cases, rises from 61.8% to 82.7% of questions, with mean calls rising from 5.08 to 8.70. Counterexample searches and scaling-limit checks, which test explicitly encoded claims through solver-based or numerical searches and symbolic calculations, rise from 2.08 to 4.18 and from 1.58 to 3.49 mean calls per question, respectively. The latest release thus relies more on model-based review of assumptions and on specialized checks, and less on general-purpose code, a change that coincides with higher scores but whose causal effect this comparison cannot isolate.

Supplementary Note S2: related benchmarks and agent designs

Related benchmarks cover several forms of research reasoning. RealMath evaluates verifiable mathematical answers drawn from research sources[58], and FrontierScience includes expert-designed scientific subtasks[59]. Physics benchmarks span curriculum problems and research tasks, including TPBench[60], CritPt[61] and PRLBench[62]. Our study holds paper-derived questions and grading fixed across successive model releases and relates performance to recorded derivation expenditure.

Agent design can also affect performance. Previous studies examine tool interfaces[63], combinations of agent roles[64] and mechanisms for organizing and updating memory[65]. Work on tool-assisted reasoning motivates making numerical and symbolic checks available: models can delegate calculations during reasoning[66] and use tool feedback to detect and revise errors[67]. The comparisons in our study evaluate the documented configurations; they do not isolate the contribution of each design choice.

Supplementary tables

Supplementary Table S1: Paired release contrasts under five grading aggregations.

Later versus earlier	Δ score points	Simultaneous 95% CI points	p_t Holm75	Δ full marks pp	p_{McN} Holm75
Recorded rubric aggregation					
o3 versus o1	4.08	$[-1.27, 9.43]$	0.0452	3.94	0.2340
GPT-5 versus o1	20.85	$[15.20, 26.50]$	7.10×10^{-21}	13.78	2.54×10^{-8}
GPT-5.1 versus o1	16.60	$[10.57, 22.64]$	6.52×10^{-13}	11.81	8.86×10^{-6}
GPT-5.4 versus o1	33.78	$[27.28, 40.28]$	2.89×10^{-36}	25.98	1.82×10^{-18}
GPT-5.5 versus o1	39.20	$[32.58, 45.81]$	1.99×10^{-44}	31.89	1.42×10^{-20}
GPT-5 versus o3	16.77	$[11.33, 22.21]$	5.88×10^{-16}	9.84	0.0003
GPT-5.1 versus o3	12.52	$[7.38, 17.66]$	9.07×10^{-11}	7.87	0.0058
GPT-5.4 versus o3	29.69	$[23.95, 35.44]$	4.66×10^{-36}	22.05	2.33×10^{-14}
GPT-5.5 versus o3	35.11	$[29.02, 41.21]$	3.69×10^{-42}	27.95	9.81×10^{-18}
GPT-5.1 versus GPT-5	-4.24	$[-9.21, 0.72]$	0.0452	-1.97	1.0000
GPT-5.4 versus GPT-5	12.93	$[7.07, 18.78]$	6.58×10^{-9}	12.20	0.0004
GPT-5.5 versus GPT-5	18.35	$[12.17, 24.53]$	1.35×10^{-14}	18.11	3.45×10^{-8}
GPT-5.4 versus GPT-5.1	17.17	$[11.21, 23.14]$	4.30×10^{-14}	14.17	1.53×10^{-5}
GPT-5.5 versus GPT-5.1	22.59	$[16.64, 28.54]$	3.08×10^{-22}	20.08	9.02×10^{-10}
GPT-5.5 versus GPT-5.4	5.42	$[0.65, 10.19]$	0.0117	5.91	0.2168
Mean of the first three votes					
o3 versus o1	4.44	$[-0.73, 9.62]$	0.0452	2.76	0.6460
GPT-5 versus o1	20.97	$[15.51, 26.42]$	2.65×10^{-22}	11.81	2.91×10^{-6}
GPT-5.1 versus o1	16.65	$[10.79, 22.51]$	1.16×10^{-13}	11.42	1.53×10^{-5}
GPT-5.4 versus o1	33.73	$[27.39, 40.06]$	1.60×10^{-37}	24.41	4.37×10^{-16}
GPT-5.5 versus o1	39.37	$[32.89, 45.86]$	7.39×10^{-46}	31.50	2.73×10^{-20}
GPT-5 versus o3	16.52	$[11.60, 21.44]$	1.65×10^{-18}	9.06	0.0004
GPT-5.1 versus o3	12.21	$[7.32, 17.10]$	2.98×10^{-11}	8.66	0.0012
GPT-5.4 versus o3	29.29	$[23.94, 34.64]$	4.55×10^{-39}	21.65	3.33×10^{-15}
GPT-5.5 versus o3	34.93	$[29.18, 40.69]$	2.60×10^{-45}	28.74	2.82×10^{-19}
GPT-5.1 versus GPT-5	-4.32	$[-8.81, 0.18]$	0.0417	-0.39	1.0000
GPT-5.4 versus GPT-5	12.76	$[7.28, 18.25]$	1.08×10^{-9}	12.60	0.0003
GPT-5.5 versus GPT-5	18.41	$[12.51, 24.30]$	7.17×10^{-16}	19.69	3.67×10^{-9}
GPT-5.4 versus GPT-5.1	17.08	$[11.45, 22.70]$	1.98×10^{-15}	12.99	6.09×10^{-5}
GPT-5.5 versus GPT-5.1	22.72	$[17.01, 28.43]$	4.19×10^{-24}	20.08	9.02×10^{-10}
GPT-5.5 versus GPT-5.4	5.64	$[1.05, 10.24]$	0.0051	7.09	0.0440
Median of the first three votes					
o3 versus o1	4.94	$[-0.45, 10.33]$	0.0417	3.54	0.3164
GPT-5 versus o1	21.69	$[16.03, 27.35]$	3.06×10^{-22}	14.17	1.46×10^{-9}
GPT-5.1 versus o1	17.23	$[11.16, 23.30]$	1.35×10^{-13}	11.42	1.53×10^{-5}
GPT-5.4 versus o1	34.66	$[28.19, 41.12]$	6.41×10^{-38}	25.59	3.58×10^{-18}
GPT-5.5 versus o1	39.82	$[33.23, 46.41]$	1.73×10^{-45}	31.50	2.73×10^{-20}
GPT-5 versus o3	16.75	$[11.46, 22.04]$	9.97×10^{-17}	10.63	4.48×10^{-5}
GPT-5.1 versus o3	12.29	$[7.22, 17.36]$	1.12×10^{-10}	7.87	0.0058

Continued on next page

Supplementary Table S1 continued

Later versus earlier	Δ score points	Simultaneous 95% CI points	p_t Holm75	Δ full marks pp	p_{McN} Holm75
GPT-5.4 versus o3	29.72	$[24.14, 35.30]$	1.00×10^{-37}	22.05	2.33×10^{-14}
GPT-5.5 versus o3	34.88	$[28.98, 40.79]$	9.04×10^{-44}	27.95	9.81×10^{-18}
GPT-5.1 versus GPT-5	-4.46	$[-9.10, 0.19]$	0.0417	-2.76	1.0000
GPT-5.4 versus GPT-5	12.97	$[7.30, 18.63]$	1.86×10^{-9}	11.42	0.0012
GPT-5.5 versus GPT-5	18.13	$[12.07, 24.19]$	8.13×10^{-15}	17.32	2.29×10^{-7}
GPT-5.4 versus GPT-5.1	17.43	$[11.69, 23.16]$	2.12×10^{-15}	14.17	1.53×10^{-5}
GPT-5.5 versus GPT-5.1	22.59	$[16.80, 28.38]$	3.58×10^{-23}	20.08	9.02×10^{-10}
GPT-5.5 versus GPT-5.4	5.16	$[0.55, 9.77]$	0.0125	5.91	0.2168
Mean of all recorded votes					
o3 versus o1	4.20	$[-0.92, 9.33]$	0.0452	2.76	0.6460
GPT-5 versus o1	21.02	$[15.57, 26.48]$	2.03×10^{-22}	11.81	2.91×10^{-6}
GPT-5.1 versus o1	16.48	$[10.67, 22.30]$	1.26×10^{-13}	11.42	1.53×10^{-5}
GPT-5.4 versus o1	33.58	$[27.22, 39.94]$	3.74×10^{-37}	24.41	4.37×10^{-16}
GPT-5.5 versus o1	39.05	$[32.57, 45.53]$	2.26×10^{-45}	31.50	2.73×10^{-20}
GPT-5 versus o3	16.82	$[11.85, 21.78]$	8.04×10^{-19}	9.06	0.0004
GPT-5.1 versus o3	12.28	$[7.39, 17.18]$	2.26×10^{-11}	8.66	0.0012
GPT-5.4 versus o3	29.38	$[23.99, 34.77]$	7.63×10^{-39}	21.65	3.33×10^{-15}
GPT-5.5 versus o3	34.85	$[29.02, 40.68]$	2.28×10^{-44}	28.74	2.82×10^{-19}
GPT-5.1 versus GPT-5	-4.54	$[-9.04, -0.03]$	0.0326	-0.39	1.0000
GPT-5.4 versus GPT-5	12.56	$[7.06, 18.07]$	2.06×10^{-9}	12.60	0.0003
GPT-5.5 versus GPT-5	18.03	$[12.13, 23.92]$	2.27×10^{-15}	19.69	3.67×10^{-9}
GPT-5.4 versus GPT-5.1	17.10	$[11.46, 22.74]$	1.98×10^{-15}	12.99	6.09×10^{-5}
GPT-5.5 versus GPT-5.1	22.57	$[16.89, 28.24]$	3.77×10^{-24}	20.08	9.02×10^{-10}
GPT-5.5 versus GPT-5.4	5.47	$[0.87, 10.06]$	0.0071	7.09	0.0440
Median of all recorded votes					
o3 versus o1	4.99	$[-0.40, 10.37]$	0.0417	3.94	0.2340
GPT-5 versus o1	21.43	$[15.75, 27.11]$	8.42×10^{-22}	14.17	1.46×10^{-9}
GPT-5.1 versus o1	16.83	$[10.83, 22.83]$	2.05×10^{-13}	11.42	1.53×10^{-5}
GPT-5.4 versus o1	34.58	$[28.07, 41.10]$	1.60×10^{-37}	26.77	4.61×10^{-19}
GPT-5.5 versus o1	38.98	$[32.40, 45.56]$	2.10×10^{-44}	31.50	2.73×10^{-20}
GPT-5 versus o3	16.45	$[11.14, 21.75]$	3.50×10^{-16}	10.24	0.0002
GPT-5.1 versus o3	11.85	$[6.77, 16.93]$	5.27×10^{-10}	7.48	0.0140
GPT-5.4 versus o3	29.60	$[23.99, 35.20]$	4.22×10^{-37}	22.83	6.14×10^{-15}
GPT-5.5 versus o3	33.99	$[28.00, 39.99]$	2.95×10^{-41}	27.56	1.22×10^{-16}
GPT-5.1 versus GPT-5	-4.60	$[-9.30, 0.10]$	0.0384	-2.76	1.0000
GPT-5.4 versus GPT-5	13.15	$[7.53, 18.77]$	9.49×10^{-10}	12.60	0.0003
GPT-5.5 versus GPT-5	17.55	$[11.46, 23.63]$	6.02×10^{-14}	17.32	2.29×10^{-7}
GPT-5.4 versus GPT-5.1	17.75	$[11.99, 23.51]$	8.45×10^{-16}	15.35	3.23×10^{-6}
GPT-5.5 versus GPT-5.1	22.15	$[16.39, 27.90]$	8.05×10^{-23}	20.08	9.02×10^{-10}
GPT-5.5 versus GPT-5.4	4.40	$[-0.26, 9.05]$	0.0417	4.72	0.5220

Each contrast subtracts the earlier release from the later release. Each displayed p-value column has its own Holm adjustment over 75 tests, using two-sided paired t tests for mean scores and two-sided exact McNemar tests for full marks. Full-mark changes are percentage-point differences, not changes in the mean score. The

source file also reports unadjusted tests, sign and Wilcoxon tests, Bonferroni intervals, gains, losses and ties. Source: `results/grading/release_contrasts.csv`.

Supplementary Table S2: Assessed candidates and included questions by field and sampling round.

	Round 1		Round 2		Included
Field	n/N	Eligible % (Wilson 95% CI)	n/N	Eligible % (Wilson 95% CI)	R1+R2
Analysis of PDEs	34/35	97.1 [85.5, 99.5]	—	—	34
Probability	33/35	94.3 [81.4, 98.4]	—	—	33
Statistics theory	34/35	97.1 [85.5, 99.5]	—	—	34
Economic theory	29/35	82.9 [67.3, 91.9]	10/10	100.0 [72.2, 100.0]	39
High-energy theory	27/35	77.1 [61.0, 87.9]	13/19	68.4 [46.0, 84.6]	40
Gravitation and cosmology	20/35	57.1 [40.9, 72.0]	17/35	48.6 [33.0, 64.4]	37
Quantum physics	20/35	57.1 [40.9, 72.0]	17/35	48.6 [33.0, 64.4]	37
Total	197/245	80.4 [75.0, 84.9]	57/99	57.6 [47.7, 66.8]	254

Here N is the number of assessed candidates and n the number included. Dashes indicate fields not supplemented in Round 2. PDEs, partial differential equations. Source: `results/sampling/round_rates.csv`.

Supplementary Table S3: Exploratory sensitivity to excluding the December 2025 posting cohort.

Model	n	Mean score (95% CI)	Full marks	Full-mark rate, % (95% CI)
All questions				
GPT-5.4	254	63.71 [59.80, 67.64]	73	28.74 [23.62, 34.25]
GPT-5.5	254	69.13 [65.31, 72.92]	88	34.65 [28.74, 40.55]
Excluding December 2025				
GPT-5.4	204	63.66 [59.29, 67.99]	62	30.39 [24.51, 36.76]
GPT-5.5	204	69.97 [65.79, 74.09]	75	36.76 [30.39, 43.14]
December 2025 only (descriptive)				
GPT-5.4	50	63.90 [55.72, 71.87]	11	22.00 [12.00, 32.00]
GPT-5.5	50	65.71 [57.15, 73.80]	13	26.00 [16.00, 38.00]

Paired GPT-5.5 minus GPT-5.4 contrasts.

	All 254 questions	Excluding December ($n = 204$)
Mean-score difference, points	5.42	6.30
Marginal 95% CI	[2.34, 8.57]	[2.78, 9.88]
Full-mark difference, pp	5.91	6.37
Marginal 95% CI	[1.57, 10.24]	[1.47, 11.76]
Full-mark gains	25	22
Full-mark losses	10	9
Retained full marks	63	53
Raw paired t p	0.0009	0.0007
Raw exact McNemar p	0.0167	0.0294
Adjusted paired t p	0.0027 (Holm15)	0.0014 (Holm2)
Adjusted exact McNemar p	0.05002 (Holm15)	0.0294 (Holm2)

The exclusion was performed after the primary analysis, using existing responses and scores. All intervals in this table are marginal two-sided 95% percentile intervals from 20,000 paired question resamples within fields, with each cohort's field counts fixed and seed 20260908. The primary p values retain the Holm adjustment across 15 release pairs, whereas the Holm adjustment for the exclusion covers two endpoints, the mean score and the full-mark rate. The December-only rows are descriptive and are not a between-cohort interaction test. Sources: `results/sensitivity/cutoff_cohort_summary.csv` and `results/sensitivity/cutoff_cohort_contrasts.csv`.

Supplementary Table S4: Mean scores, full marks and question-level changes by scientific field.

Field	n	o1 mean (%)	GPT-5.5 mean (%)	Mean change (points)	GPT-5.5 full marks	Improved / unchanged / decreased	Slope (points/ month)
Analysis of PDEs	34	28.9	68.2	+39.3	8	26 / 4 / 4	+2.38
Probability	33	35.1	74.2	+39.1	16	28 / 2 / 3	+2.49
Statistics theory	34	38.1	77.5	+39.4	14	28 / 3 / 3	+2.28
Economic theory	39	30.8	67.9	+37.1	14	27 / 8 / 4	+2.22
High-energy theory	40	16.0	57.9	+42.0	9	37 / 1 / 2	+2.54
Gravitation and cosmology	37	22.0	63.3	+41.3	9	32 / 3 / 2	+2.61
Quantum physics	37	41.0	77.0	+36.1	18	29 / 6 / 2	+2.00
Total	254	29.9	69.1	+39.2	88	207 / 27 / 20	+2.36

Mean changes and improved/unchanged/decreased counts compare GPT-5.5 with o1, question by question. Slopes are fitted linear trends over all six release dates, expressed per month using $365.25/12$ days per month, and are not the endpoint difference divided by time. Total counts are sums, and total means and the total slope weight fields by their numbers of questions. Displayed differences are calculated before rounding. PDEs, partial differential equations. Source: `results/performance/field_validation.csv`.

Supplementary Methods: prompt and implementation index

Supplementary Code 1 includes nine prompt templates (31,993 words) with standardized stage terminology at their production locations under `prompts/curation/`, `prompts/derivation/` and `prompts/evaluation/`. Seven indexed Python files document message assembly and dispatch in the same production tree. They show how templates are combined with item content, discipline placeholders, tool-policy overlays and the conditional same-model advisory. Each file is included once. Paths below are relative to the repository root, which contains both the code and data required to run the offline analyses.

These materials are templates and implementation files, not full instantiated conversations for every question or a complete transcript of every API request. The retained conditional prompts support Curation repair and Derivation gap resolution and self-critique; these stages need not run for every answer. Configuration and outcome records must be consulted to determine which stages and settings applied. Stage terminology, identifiers and paths in these templates, implementation files and recorded text have been standardized to Curation, Derivation and Evaluation. References to the removed post-scoring stages have also been deleted from the retained prompts, while preserving direct answer-quality requirements. These packaging changes do not represent a new model experiment and leave the recorded numerical outcomes unchanged. SHA-256 hashes below identify the packaged files; `prompts/manifest.json` also preserves the hashes recorded before this migration, alongside normalized source paths, sizes and scopes.

Recorded-analysis scope. The current implementation contains Curation, Derivation and the three-component Evaluation. Separate post-scoring audits were performed in some historical runs; their penalties and model calls are outside the reported scoring and cost analysis. Supplementary Data 1 contains the canonical scores, selected answers, grading votes and published cost ledger. Full event logs for those additional audits are not included. The compact scope record in `docs/provenance.json` documents this boundary.

Prompt templates with standardized terminology

P01. Curation auditor `prompts/curation/auditor_prompt.txt` (36,278 bytes).

Instructions for auditing the constructed question, premises and hidden grading material.

SHA-256: `14827831beef4f88b243713bb585032e88e09b1207c60cea2fe2488182b5f1a4`

P02. Curation question construction `prompts/curation/benchmark_prompt.txt` (62,937 bytes).

Instructions for constructing the public input and hidden grading material from a source paper.

SHA-256: `1a94c64c3a8042138e2e513646b76a3e405207bdcb5cffeef4605bcf1e7b0bdb`

P03. Curation repair `prompts/curation/repair_prompt.txt` (21,924 bytes).

Instructions for repairing a constructed item when this stage is invoked.

SHA-256: `4828a34cda3bc8dc0b0628b492f4071ad1840eed7cd47122b2bef8e602c6d70b`

P04. Curation screening `prompts/curation/triage_prompt.txt` (12,293 bytes).

Instructions for assessing the contribution type and eligibility of a candidate paper.

SHA-256: `ca0768a4d430a9f12c491e89660cb636a555d6b6552eea617a7eeea9659143b2`

P05. Derivation gap resolution `prompts/derivation/gap_resolver.txt` (6,381 bytes).

Instructions for the optional gap-resolution stage; presence in the archive does not establish use for every outcome.

SHA-256: `f78550828e1eb7563e6ee914777d8d2423adcd2d2f6c2584406921e0309f246e`

P06. Derivation answer generation `prompts/derivation/derivation.txt` (60,186 bytes).

Base instructions for generating a response; runtime construction substitutes the discipline and can alter tool guidance.

SHA-256: `a47d3b6e32a2b74d763c21f15c061706204a9c3eb6f0837a68b8bc0a05694756`

P07. Derivation self-critique `prompts/derivation/self_critique.txt` (23,194 bytes).

Instructions for internal critique of a generated draft when invoked.

SHA-256: `90e22829a8d3e8e3b0e3ce072db36dcacfd4952293e30860058bbd8e835f8130`

P08. Derivation revised emission `prompts/derivation/self_critique_emit.txt` (6,888 bytes).

Instructions for emitting a revised draft after internal critique when invoked.

SHA-256: `776e681fb3054b5cccf87a9a4bde4294658f7b655d74fde1337cd6fb2e0e8913`

P09. Evaluation grading `prompts/evaluation/evaluation.txt` (35,764 bytes).

Base instructions for rubric-based grading; item and response content are assembled separately.

SHA-256: `6ee383bf45845becf88db404eb1cd8782a31ff8690e28eedc94dd1809dc83a8a`

Assembly and dispatch source

C01. Curation assembly `pipeline/curation/prompting.py`.

Loads Curation prompts and constructs stage-specific messages, including source metadata, fetching instructions and mechanical reminders.

SHA-256: `18402576e5bff907f4d4439b3c9696b369d03e14a0182831e19ea88da8f470a9`

C02. Derivation input assembly `pipeline/derivation/prompting.py`.

Loads Derivation templates and constructs the question-and-premises text presented as public input.

SHA-256: 73a1c8d409e0b1c0e17f3bee10b962c8aecbf235c68a17b6762526f799863125

C03. Derivation instruction overlays `pipeline/derivation/agents/factories.py`.

Substitutes discipline placeholders, applies the selected tool-policy section and appends category guidance when configured; registers the available tools.

SHA-256: 33295b8cdc58c10b7cb9b213775580b0ede82244ffd70add88f657bf62f21a6a

C04. Category guidance `pipeline/shared/tools/derivation_prompt_guidance.py`.

Provides the category-specific guidance appended by the Derivation agent factory under the configured track.

SHA-256: 3100f8ea6987bc7a48d016309ec71b788639d0913dd9e0720722b1c2dadcbc14

C05. Tool catalog `pipeline/shared/tools/catalog.py`.

Defines the canonical registration order and specifications for scientific checking tools.

SHA-256: 236cb0010daff48d2155d91c611b78a11f61314c9e7e642b15d4052caacc9b5b

C06. Evaluation message assembly `pipeline/evaluation/prompting.py`.

Constructs grading context from metadata, public input, hidden grading material and response; appends the same-model advisory when the flag is true.

SHA-256: 4fd16aaed98ab4d945d658daf7c91a2554b8764953666839aa66b2c497cb6798

C07. Evaluation dispatch `pipeline/evaluation/pipeline/stages.py`.

Computes the same-model flag by comparing configured Derivation and Evaluation model names, passes it to the message builder and incorporates the condition into grading-cache keys.

SHA-256: f1875edb0dafe48a4499e535c5ba9990f5fc64ace233940e3ba302a8e928dd50

Implementation of the same-model advisory. The Evaluation dispatcher sets the flag when the configured Derivation and Evaluation model names match after trimming whitespace. The message builder then appends the advisory immediately after the answer produced by Derivation. The advisory does not change the scoring formula.

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